BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Compilers and Interpreters

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why is a translator needed for high-level source code?

2. Which type suits a count of pupils?
   A) Real
   B) Boolean
   C) Character
   D) Integer

3. "A character data type can hold an entire arbitrary-length sentence as one character." Correct this. Explain why.

4. Start with unsigned 00000000. Set only the MSB to 1, then only the LSB to 1. Give both denary values. Why do they differ?

Today's objectives

Explain why translators are needed. Trace the compilation and interpretation processes.

Key vocabulary - Read, cover, recall

Translator
Program that converts source code into a form a computer can execute.

Compiler
Translator that processes the whole source program before execution and produces an executable.

Interpreter
Translator that executes a program as it translates it during running.

Source code
Instructions written by the programmer in a programming language.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Why translation is needed

A processor runs machine instructions. A translator converts source code written in a programming language so the computer can execute it.

Compiler

A compiler translates the entire program before it is run and produces executable code. The executable can normally be run again without recompiling unchanged source.

Interpreter

An interpreter translates and executes the program during running, commonly one statement at a time. It usually has to be present each time the source is run.

Choose and compare

Compilation is useful when a finished program runs repeatedly or is distributed. Interpretation can be useful when developing and testing; error feedback can appear at the statement being executed.

I Do - Worked model

Explain TWO features of a compiler. [4 marks]

- A programmer begins with source code: this is not normally directly executable by the CPU.
- The compiler translates the whole source program before it runs [1], rather than executing each statement as it is processed [1].
- The compiler produces executable machine code [1] that the CPU can run.
- The executable can be run repeatedly without translating the unchanged source again [1].

We Do - Guided practice

1. Explain TWO reasons an interpreter might be useful while developing a program. [4 marks]

2. Compare when a compiler and an interpreter translate code, and what happens when the program runs again. [4 marks]

3. An application is distributed to many school computers and run every day. Explain TWO benefits of compiling it before distribution. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the missing closing bracket in this character-to-code conversion.

```
print(ASC("B")
```

You Do - Independent exam questions

Q2. A compiler reports an error before the programmer can run a new program. Explain why translation is necessary and how the compiler helps find a problem. [4 marks]

Q3. A learner changes the source code of a small program many times while testing. Explain ONE benefit and ONE drawback of interpreting the program. [4 marks]

Q4. A game studio wants users to run a finished game repeatedly on their own computers. Explain TWO advantages of compiling the game. [4 marks]

Exit Ticket - True or False?

A compiler normally translates the whole source program before the executable runs.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. A CPU runs machine instructions. High-level source code must be translated before or as it runs.

2. D - Integer

3. A character represents one character; a string represents a sequence of characters.

4. MSB only: 128. LSB only: 1. The difference is 127.

I Do - Explain the model

A programmer begins with source code: this is not normally directly executable by the CPU.

The compiler translates the whole source program before it runs [1], rather than executing each statement as it is processed [1].

The compiler produces executable machine code [1] that the CPU can run.

The executable can be run repeatedly without translating the unchanged source again [1].

We Do - Model answers

1. It translates and executes during running [1], allowing source code to be tested without producing a separate executable [1]. Errors can be reported at the statement reached [1], helping the programmer locate a fault [1].

2. A compiler translates the whole program before execution [1]; an interpreter translates and executes during running [1]. A compiled executable can normally run again without recompilation [1]; an interpreted program normally needs the interpreter each time [1].

3. The compiled executable can be distributed to run on the target platform [1], without each user needing the source-language interpreter [1]. The unchanged executable can be run many times [1], without repeating compilation on every launch [1].

Code First - Answer

print(ASC("B"))
Expected output: 66

You Do - Self-mark

Q2: The CPU needs executable machine instructions [1], so source code must be translated [1]. The compiler attempts to translate the whole source [1] and reports errors found during compilation so the programmer can correct them [1].

Q3: An interpreter translates and executes during running [1], allowing the learner to test changed code without a separate compilation step [1]. The interpreter is required at each run [1], which may make repeated execution slower than running a compiled executable [1].

Q4: The whole program is translated into executable code [1], so the distributed version can run on the target system [1]. The executable can run repeatedly [1], so recompilation of unchanged source is not needed each time [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.